

PROJECT FINGER PRINTER

1st PHASE PRESENTATION

GROUP B

Group Members : Mithin Kumar Chithirala

Madana Gopala Reddy Konda

Yuvraj Ramchandra Labhade

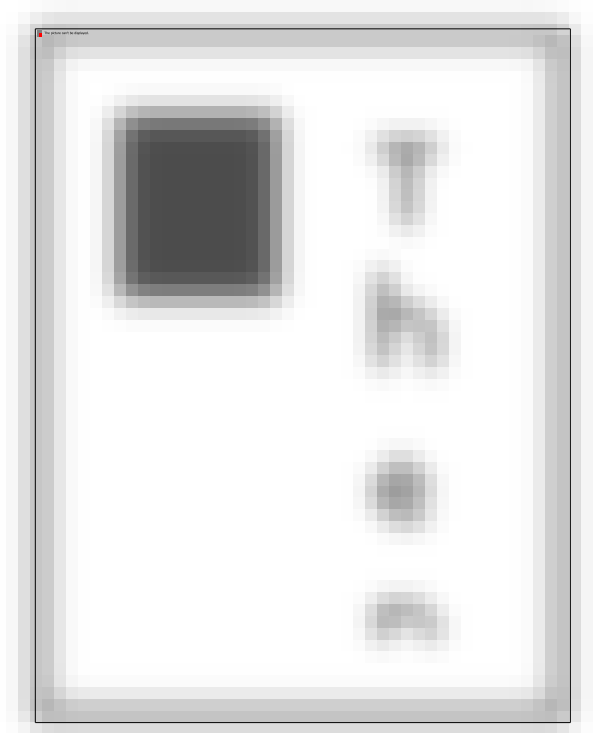
Vinay Kumar Pepeti

Outline

1. Getting familiar with 3D printer
2. Concept Overview
3. Preliminary CAD Model
4. Actuation System Design
5. Procurement Strategy: Make vs. Buy
6. Concept Evaluation & Performance Comparison
7. Optimization Strategies

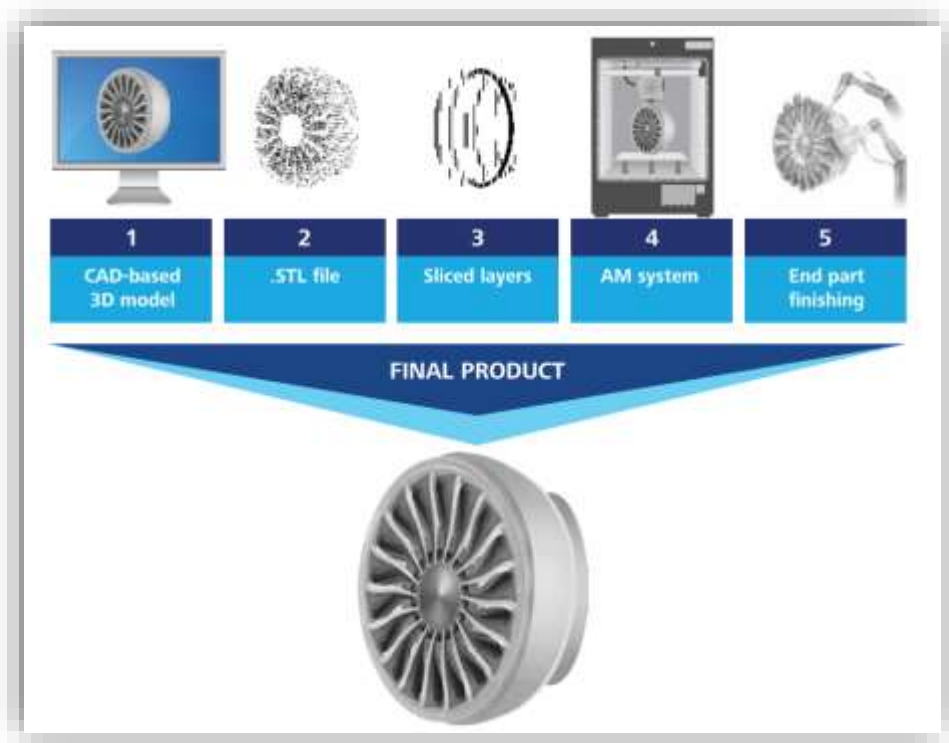
1. Getting familiar with 3D printer

- Fused Deposition Modeling (FDM), or Fused Filament Fabrication (FFF), is a cost-effective 3D printing method that creates objects by layering melted thermoplastic filament.
- **Fundamental Principle:** FDM works by heating and extruding thermoplastic filament layer by layer, with each layer bonding to the previous one as the nozzle moves upward to build the object.



<https://blog.goldsupplier.com/p/arts-of-a-3d-printer/>

3D Printing Workflow



<https://www.swms.de/caesa/layers-and-slicing/#chfy-coo-check-save>

Stages of the 3D Printing Process

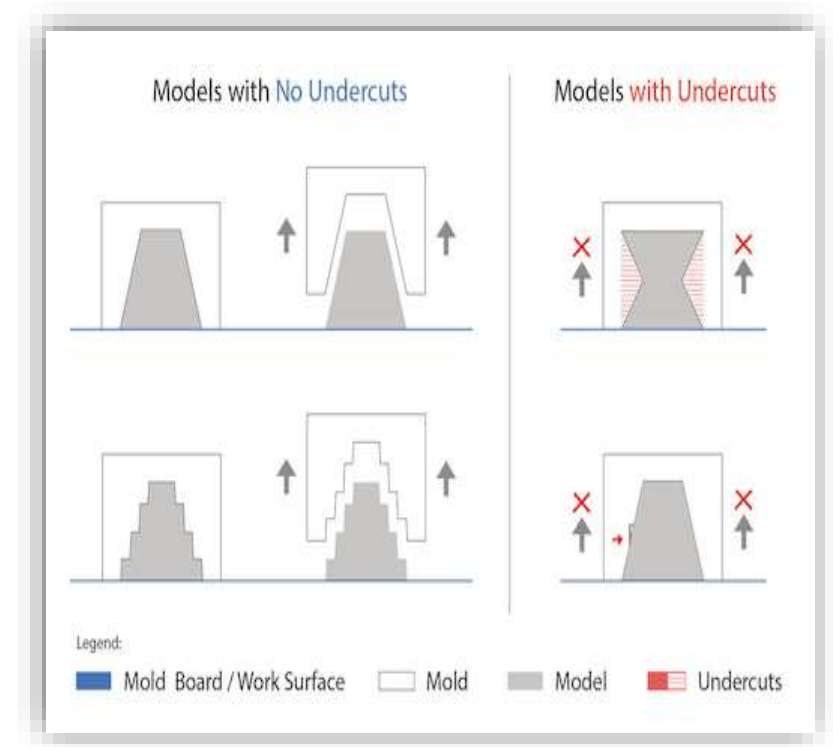
1. CAD Model
2. STL
3. Slicing 3D Printing
4. AM Systems
5. Post-Processing

Problem definition and goals

- **Overhang:** A part feature that extends outward beyond 45°, often requiring support in FDM printing.
- **Undercut:** A recessed area or groove that cannot be reached from above, making it hard to print without special mechanisms.

Project Goals

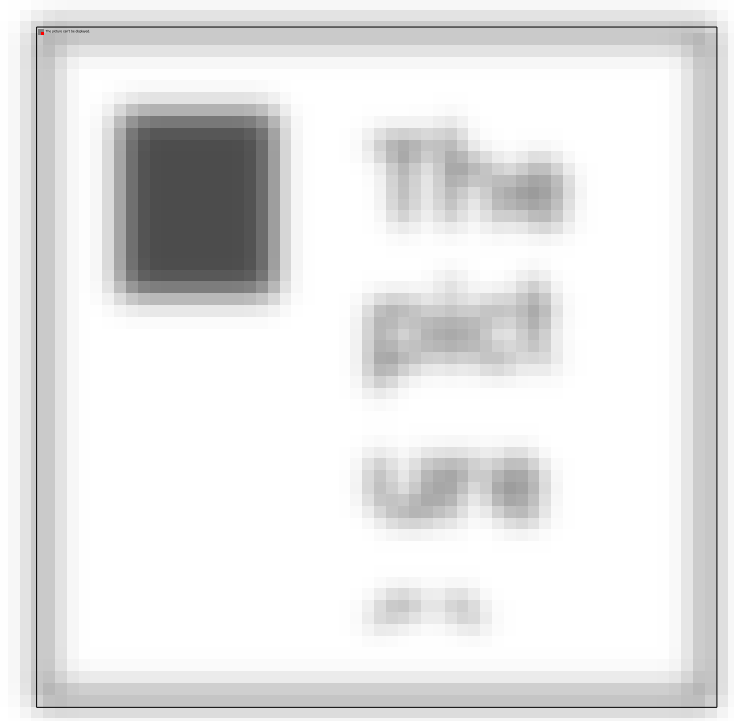
- Develop a fingerprint tool mount for 3D printer integration.
- Enable 5-axis movement for better reach, precision, and surface finish.
- Optimize for complex shapes with minimal supports.
- Minimize weight and cost through smart design and materials.



<https://images.app.goo.gl/wsfjruP8TP9mWCDY7>

Tronxy X5SA Pro – Core Features & Capabilities

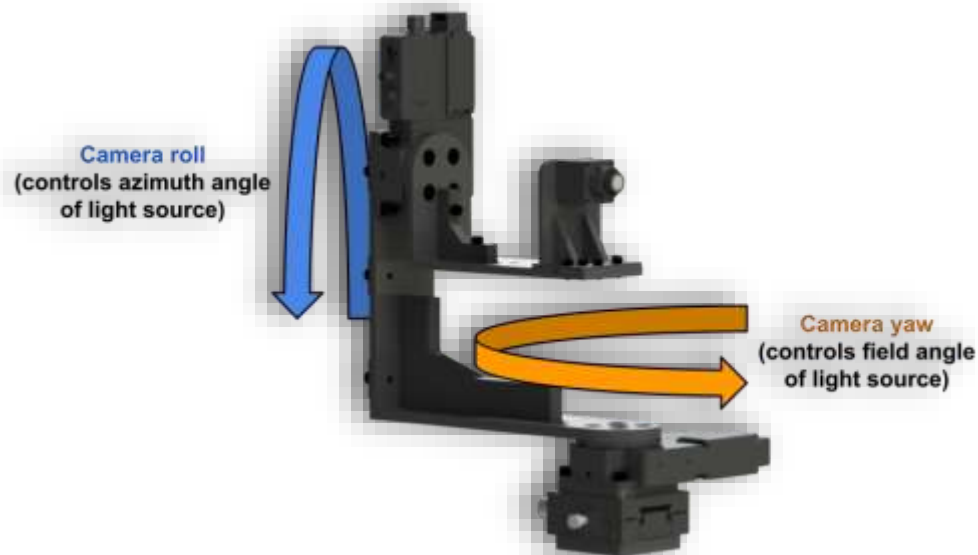
Feature	Specification
Technology	FDM (Fused Deposition Modelling)
Build Volume	330 × 330 × 400 mm
Machine Dimensions	580 × 645 × 660 mm
Nozzle Size	0.4 mm
Max Nozzle Temp	260 °C
Extruders	Single Extruder
Slicing Method	Layer-by-Layer
Print Speed	20–150 mm/s
Recommended Speed	60 mm/s



<https://www.tronxy3dprinter.com/>

2. Concept Overview

- **Idea Genesis** : Gimbal Camera
- **Key Points** : Multi-Axis Precision, Compact Design, Adaptive Printing Paths, Reduced Material Waste.



<https://www.imatest.com/docs/stray-light-flare/motorized-gimbal-instructions/>



<https://images.app.goo.gl/aJJsvNoPhp3VRS6a9>

Design Journey



3. Preliminary CAD Model

Technology : FDM (Fused Deposition Modelling)

Dimensions (X,Y,Z) : 111.5*35.98*227.8mm

Nozzle Material : 0.4mm

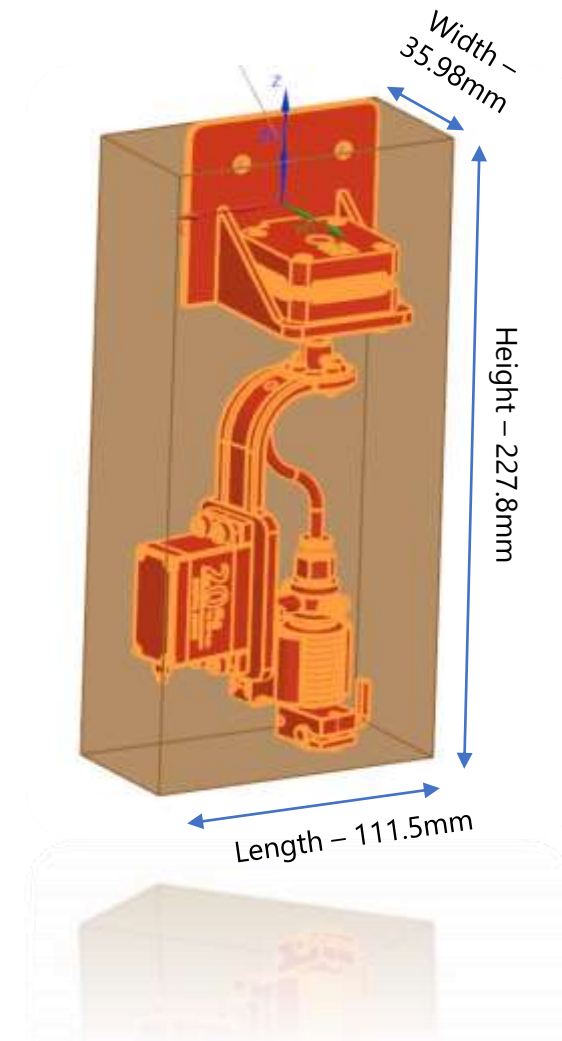
DOF : 2 DOF

Mass : 418 g

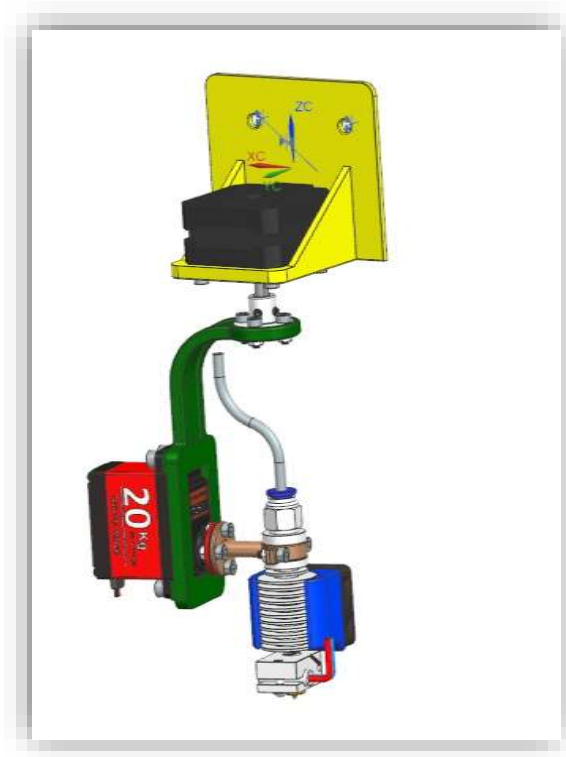
Max motion resolution : 0.7 degree

Body Material : Aluminium 6061

Flow Rate : 10mm³/Sec



Preliminary CAD Model



Kinematic Design: 2-DOF rotational system using a gimbal mechanism.

Actuation:

Servo motor for pitch (tilt) control of the print head.

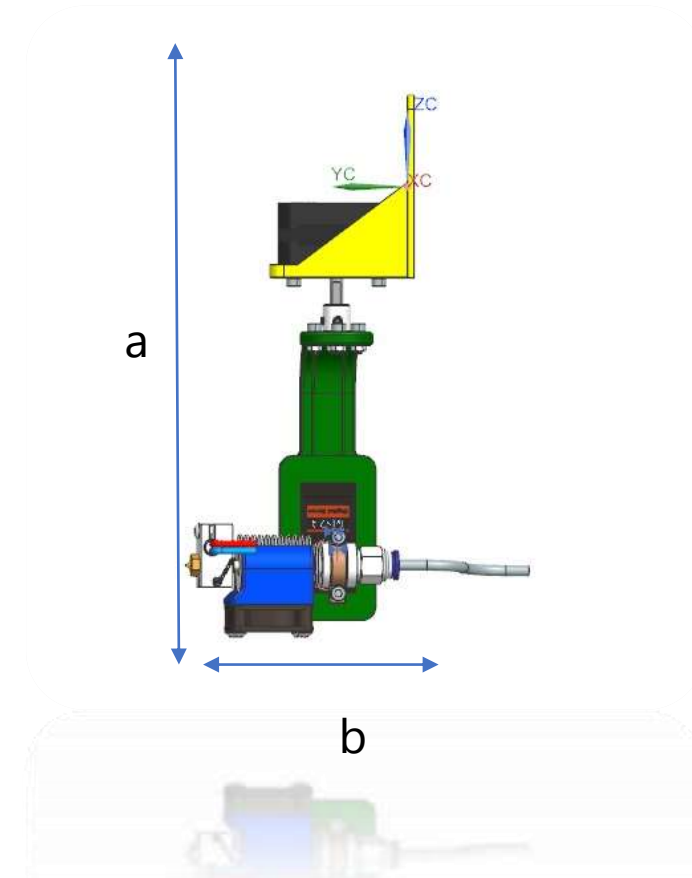
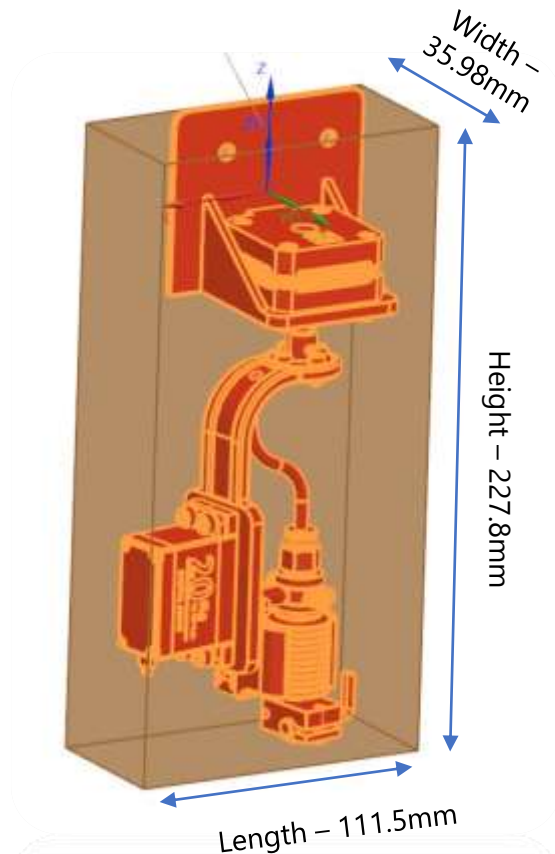
Stepper motor for rotation around the Z-axis (yaw).

Extrusion Setup: Bowden-type.

Material: Aluminum 6061

Preliminary CAD Model – General Dimension

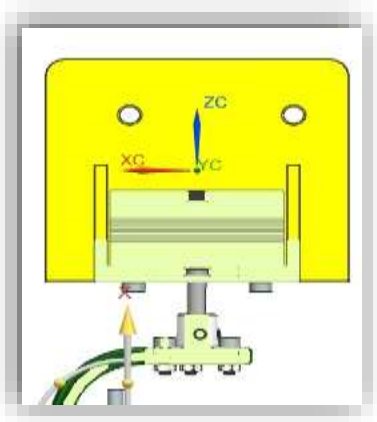
- At max. vertical position (a x b x c) : 35.98*111.5*227.8mm
- At max. horizontal position (a x b x c): 195.15*80*35.98mm



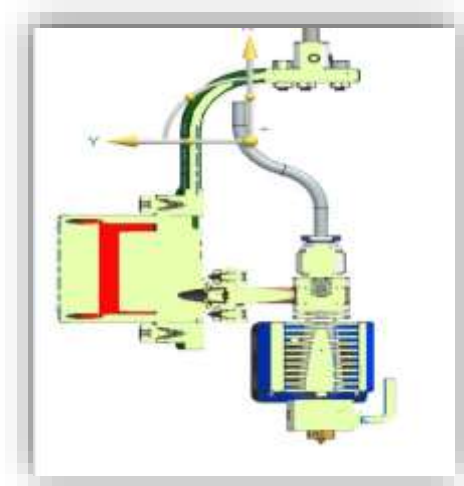
Comparative Analysis: Standard FDM Extrusion vs. Multi-Axis Nozzle Systems

Aspect	Traditional Nozzle Design	Current Nozzle Design
Degrees of Freedom (DOF)	Typically 3 DOF (X, Y, Z)	5 DOF (X, Y, Z + Orientation: Pitch, Yaw)
Overhang Capability	Limited; requires support structures	Excellent; can tilt nozzle to build overhangs without supports
Undercut Feasibility	Practically impossible without supports	Possible due to rotational nozzle head orientation
Print Surface Accessibility	Only top-down	Can access sides, angled planes
Nozzle Orientation	Fixed	Adjustable (dynamic during print)
Material Deposition Control	Linear & limited	Angular & precise
Toolpath Optimization	Less efficient for 3D contours	Adaptive to geometry
Part Strength	Anisotropic, due to Z-layer buildup	Improved due to multi-directional deposition

Preliminary CAD Model – Packing Concept

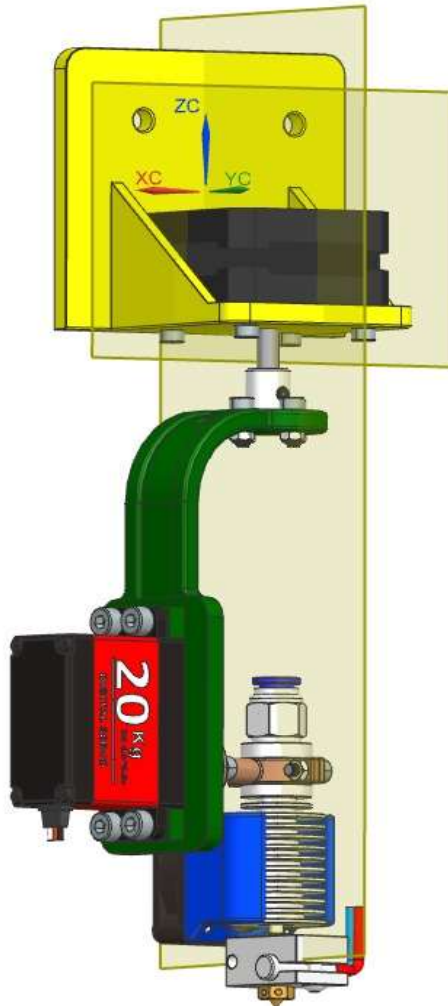


- Main assembly consists of two assemblies
Rotation Assy (Upper part).
Gimbal Assy (Lower part).



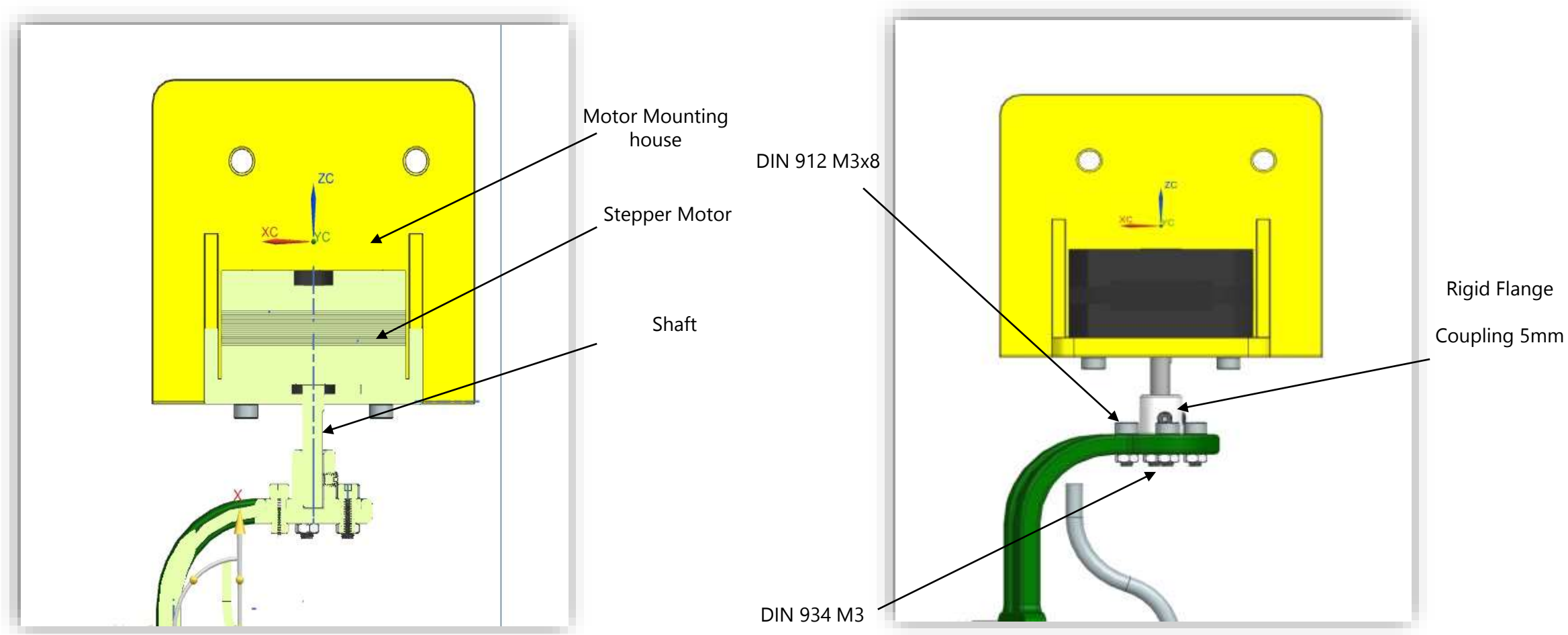
- Referenced to the same skeleton sketch.
Connected through skeleton.
Ensures parallel design.

Preliminary CAD Model – Packing Concept

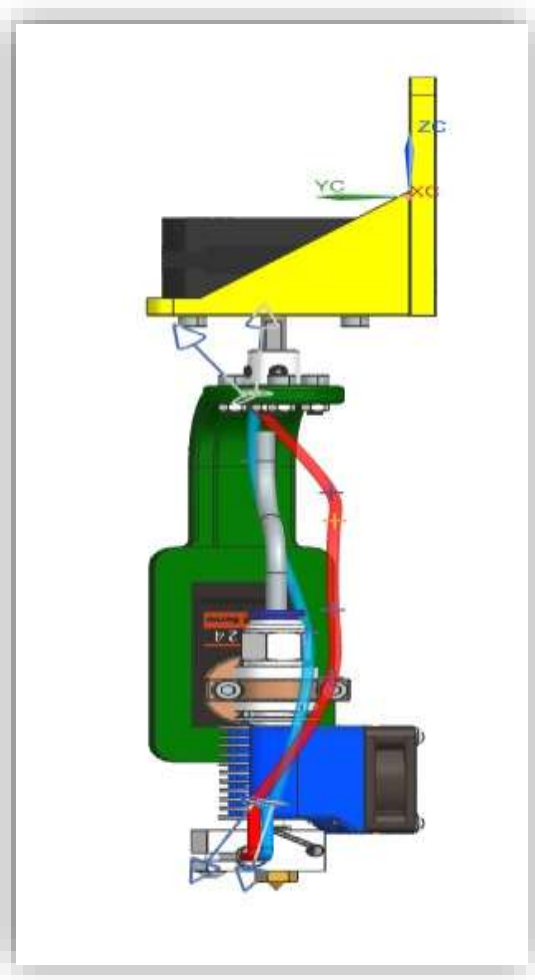


- Skeleton sketch defines;
 1. Joint position for gimbal mechanism.
 2. Z-axis rotation position .
 3. Boundary box dimension for housing & gimbal mechanism.

Preliminary CAD Model – Packing Concept



Preliminary CAD Model – Wiring Concept



Designed as a flexible component that adapts its shape based on positional changes.

4. Actuation System Design – DOF and Joints

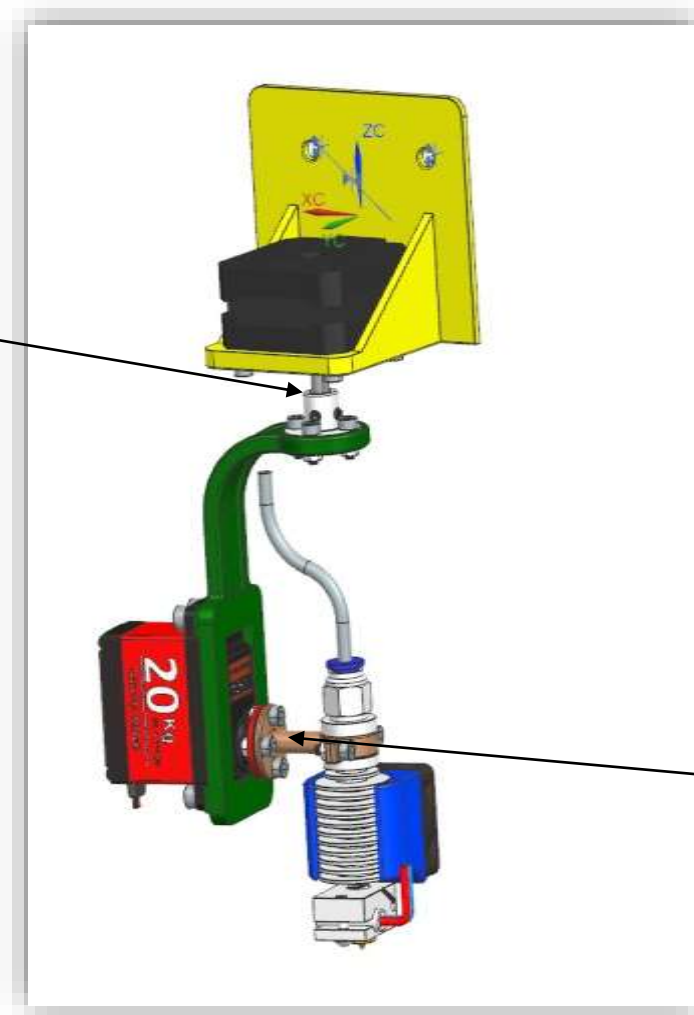
Joint 1

Type - Revolute Joint

Rotation Axis- Z Axis

Rotation limit - 360°

Total DOF - 2



Joint 2

Type-Revolute Joint

Rotation limit - 270°

Actuation System Design – Representation of DOF



Actuation System Design

NEMA 17 Stepper Motor

Parameter	Specification
Purpose	Rotation of the arm in Z-axis
Motor Type	2-Phase Hybrid Stepper
Step Angle	1.8° per step (200 steps/rev)
Holding Torque	0.1 – 0.15 Nm
Frame Size	NEMA 17 (42.3 mm × 42.3 mm)
Body Length	Approx. 23 mm
Weight	132 g (0.132 kg)
Cost	€8.49



Actuation System Design

Servo DS3218 PRO

Parameter	Specification
Purpose	Rotation of the hot end
Motor Type	DS3218 PRO (High Torque Digital Servo)
Torque	23.5 kg·cm @ 6.8 V (2.3 Nm)
Speed (no load)	0.09 sec/60° @ 6.8 V
Rotation Angle	180° (standard), some variants support 270°
Dimension	40 × 20 × 40.5 mm
Weight	60 g
Cost	€11.99



5.Procurement Strategy: Pre-Manufactured (Off-the-Shelf) Components

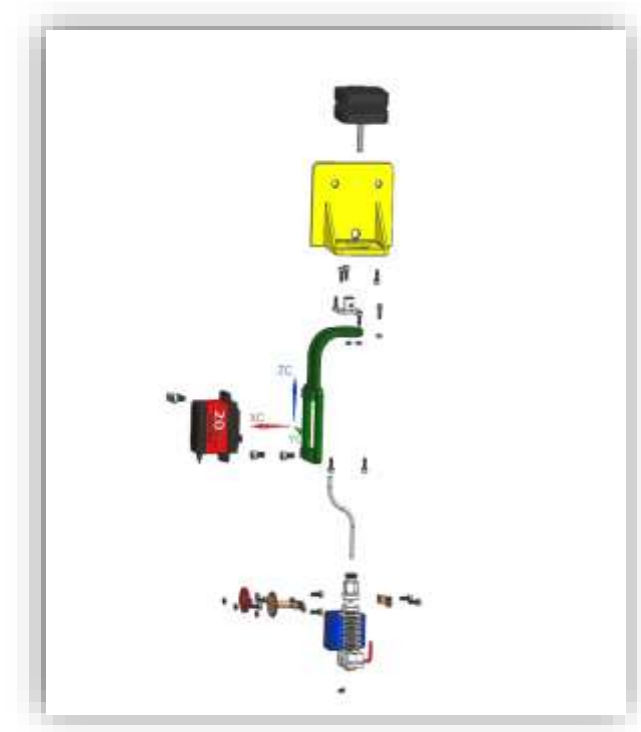
Image	Name	Price (€)	Parts Required	Total Price
	Nema 17 Motor	8,49	1	8,49
	Rigid Flange Coupling 5mm	0,99	1	0,99
	Servo DS3218 Pro	11,99	1	11,99
	Hot End V6 E3D	42,36	1	42,36
Total				63,83

Procurement Strategy: Standard Parts

Image	Name	Price (€)	Parts Required	Total Price
	DIN 912 M3x8	0,11	14	1,54
	DIN 912 M4x8	3,61	4	14,44
	DIN 934 M3	1,43	10	14,30
Total				30,28

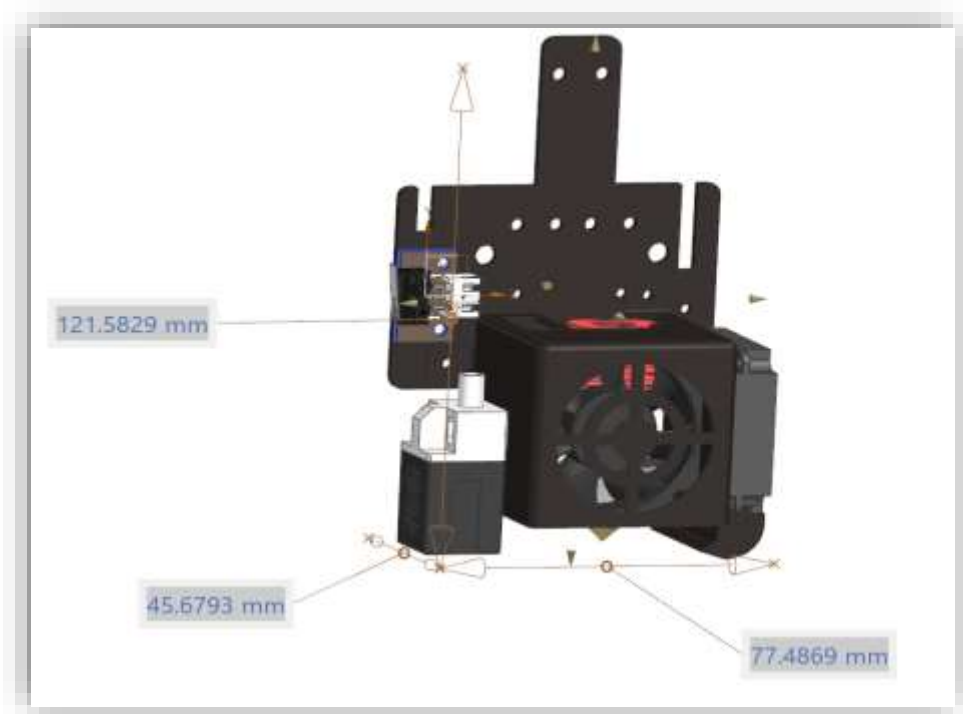
Procurement Strategy: Cost Estimation

Name	Price (€)
Off Shell Parts	63,83
Standard Parts	30,28
Manufactured Parts*	50,00
Total	144.11

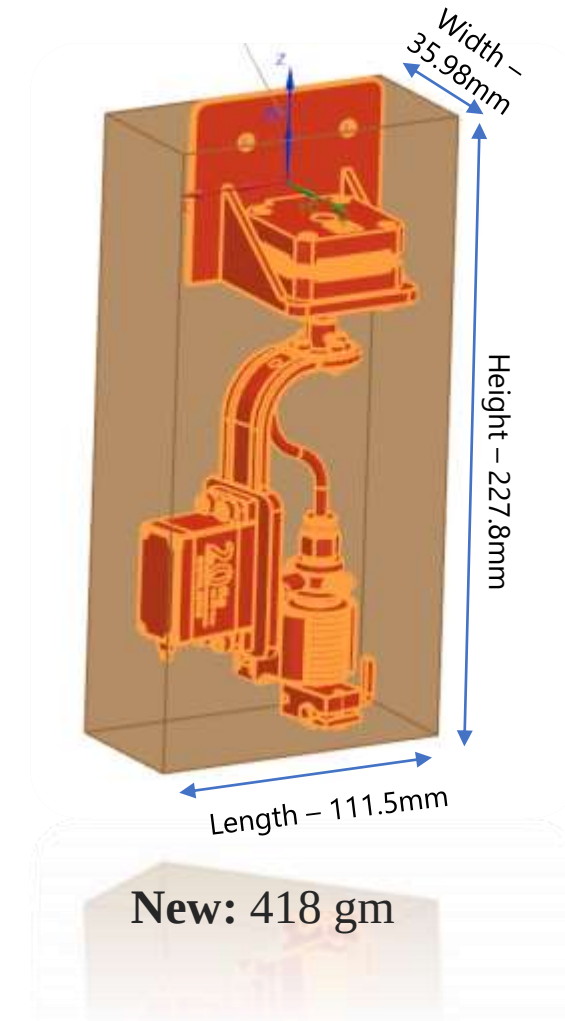


This is an estimated cost based on material and machining. Final cost may change, so manufacturer confirmation is needed.

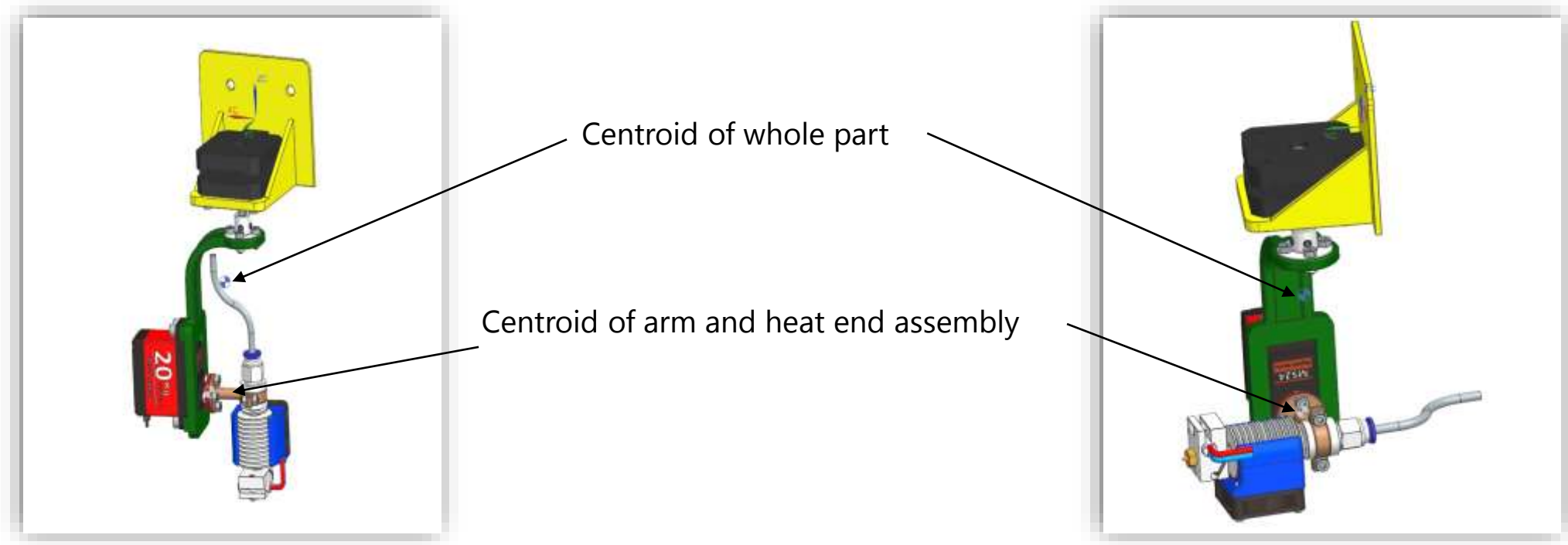
6. Concept Evaluation & Performance Comparison



Old: 382.7 gm



Concept Evaluation & Performance Comparison

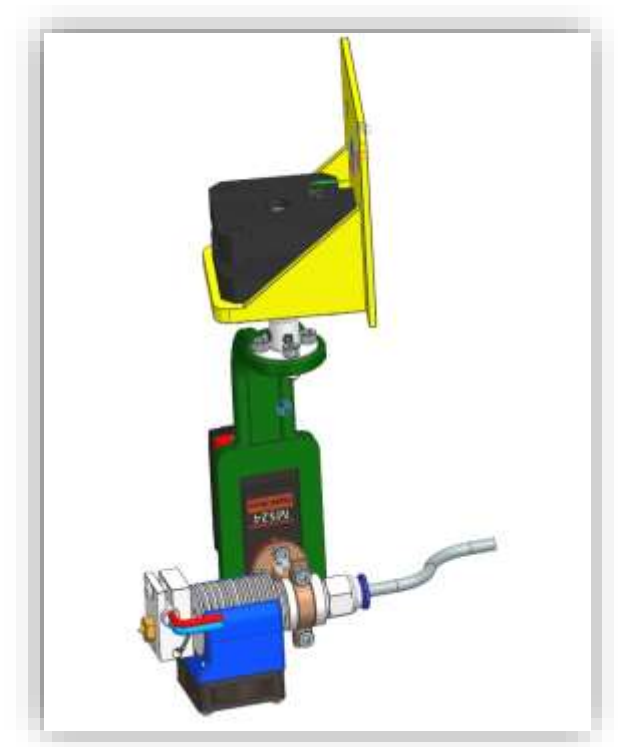


Concept Evaluation

Mass Properties Report: (All positional results with respect to Displayed Part: assembly15062025.prt: 6-16-2025 2:10)							
Component Group Used : AllComponents (Component count : 41)							
Arrangement : Arrangement 1							
Mass Properties were asserted							
Accuracy : Standard							
Root part properties:							
Density	0.00006276	kg/mm ³	Center of Mass (WCS)		First Moments (WCS)		
Area	158866.480100000	mm ²	Xcbar	13.212500000	mm	Mxc	5.522800000 kg·mm
Volume	66606.649500000	mm ³	Ycbar	24.737600000	mm	Myc	10.340300000 kg·mm
Mass	0.418000000	kg	Zcbar	-79.334100000	mm	Mzc	-33.161600000 kg·mm
Moments of Inertia (WCS)							
Ixxw	4627.814900000	kg·mm ²	Moments of Inertia (Centroidal)		Moments of Inertia (Spherical)		
Iyyw	4643.575800000	kg·mm ²	Ixx	1741.171600000	kg·mm ²		1997.858100000 kg·mm ²
Izzw	643.553600000	kg·mm ²	Iyy	1939.757200000	kg·mm ²		
			Izz	314.787500000	kg·mm ²		
Products of Inertia (WCS)							
Pyzw	-843.582300000	kg·mm ²	Products of Inertia (Centroidal)		Principal Moments of Inertia		
Pxzw	-693.373600000	kg·mm ²	Pyz	-23.241700000	kg·mm ²	Ixxp	1946.201800000 kg·mm ²
Pxyw	172.028200000	kg·mm ²	Pxz	-255.224300000	kg·mm ²	Iyyyp	1779.521500000 kg·mm ²
			Pxy	35.406200000	kg·mm ²	Izzp	269.993100000 kg·mm ²
Radii of Gyration (WCS)							
Rgxw	105.220400000	mm	Radii of Gyration (Centroidal)		Radii of Gyration (Spherical)		
Rgyw	105.399400000	mm	Rgx	64.540500000	mm		69.134400000 mm
Rgz	39.237800000	mm	Rgy	68.121700000	mm		
			Rgz	27.442300000	mm		
Principal Axes							
Xp(X)	-0.188900000		Yp(X)	0.966900000		Zp(X)	0.171300000
Xp(Y)	0.981900000		Yp(Y)	0.188800000		Zp(Y)	0.017300000
Xp(Z)	-0.015600000		Yp(Z)	0.171500000		Zp(Z)	-0.985100000

Concept Evaluation & Performance

- Mass of the product= 0.418 kg
- Estimated Torque required by the motor = 0.29kgcm
- Centre of Gravity (C.G) from reference plane :
 $X=13.21 \text{ mm}$; $Y=24.73\text{mm}$; $Z= -79.33 \text{ mm}$
- Maximum undercut= 60mm
- Maximum print angle = 270°
- Estimated Cost = 145 €



7.Optimization Strategies

- Lightweight Design**

- Use structural optimization to reduce mass without compromising rigidity.

- Compact Form Factor**

- Shrink the bounding dimensions for better space utilization and system integration.

- Joint Configuration Tuning**

- Adjust gimbal axis geometry for precise, low-friction two-axis rotation.

- Actuator Load Validation**

- Analyse motor torque via multibody simulation to ensure dynamic performance.

- Manufacturing Readiness Check**

- Review part designs for machining ease and assembly feasibility.

References

- <https://de.aliexpress.com/item/1005003897311523.html>
- [3mm 3.17mm 4mm 5mm 6mm 6.35mm 7mm 8mm 10mm 11mm 12mm Rigid Flange Coupling Motor Guide Shaft Coupler Motor Support Connector - AliExpress 13](#)
- [20 kg Servo DS3218/PRO 180 270 Grad Hochgeschwindigkeitsservo Wasserdichtes Metallgetriebe Digitalservo RC-Servo für RC-Cars im Maßstab 1/8 1/10 - AliExpress 26](#)
- https://www.3djake.de/e3d/v6-all-metal-hotend-direct-drive-175-mm?sai=7340&gad_source=1&gad_campaignid=21794339456&gbraid=0AAAAADzcMj_BtATCsg1lueGcbZSvrYXsF&gclid=CjwKCAjwgb_CBhBMEiwA0p3oOKzry6Wcx9dsvBE-cjIVnKHua9yMAudZbVTyyj1W5eN547StQcTjIRoCw5EQAvD_BwE
- https://www.fk-soehnchen.de/werkstatt/verbindungstechnik/schrauben-muttern-unterlegscheiben/schrauben/M3x8-Zylinderkopfschraube-8-8-verzinkt-DIN-912.html?srsIid=AfmBOooJ3eHGibJbrDd_FuNPxflwtCx6gipuJlid8JlcV9h4CmjHUcMe2ol
- https://schraubengrosshandel.de/inbusschrauben-m4-x-8mm-verzinkt-zyylinderkopf-din912-100-stueck?gad_source=1&gad_campaignid=22558897822&gbraid=0AAAAA_K6ZBHSalpg1HJ6SKuvsUqI_hcYX&gclid=CjwKCAjwgb_CBhBMEiwA0p3oOGj_N5hGexD0G-i5A0C3jX86Tg2Tp38h0fZwnt4H3aM7dteA53EeExoCp58QAvD_BwE
- https://schraubengrosshandel.de/sechskantmuttern-m3-din934-verzinkt-200-stueck?gad_source=1&gad_campaignid=22558897822&gbraid=0AAAAA_K6ZBHSalpg1HJ6SKuvsUqI_hcYX&gclid=CjwKCAjwgb_CBhBMEiwA0p3oOAHOG-SDAJznhgphHsuYpnsUwj0TmIT8FaEbY1RODM8ZUubBEEZOuxPBoC0E4QAvD_BwE

Thank you!